SHED MEASUREMENT WORKING PAPER

Liquid Buffers, Borrowed Resilience, and Household Hardship Profiles

A SHED measurement paper on emergency savings, financing profiles, and hardship

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Abstract

This paper develops a descriptive measurement framework for household hardship profiles in the 2025 Survey of Household Economics and Decisionmaking (SHED). The central claim is narrow: income and emergency-savings statistics are useful but incomplete because adult respondents with similar income can differ in liquid-buffer status, emergency-financing options, credit constraints, bill pressure, external support needs, and medical debt. The paper separates three objects that are often blurred together: liquid-buffer exposure, emergency-financing behavior, and hardship outcomes. It then validates an emergency-financing taxonomy that distinguishes cash/checking/savings, bridge credit paid in full, revolving credit for a $400 emergency, other emergency borrowing, inability to pay, and pooled liquidity-exclusion/unclassified states, while retaining detailed small cells as diagnostics. To reduce mechanical overlap, the primary hardship outcome is a strict non-overlap measure based only on bill pressure, external help with expenses, or medical debt; broader composites are reported as robustness outcomes. The profile decomposition is informative: strict hardship is 24.7 percent among bridge-credit-paid-in-full respondents, 31.4 percent among cash/checking/savings respondents, 67.9 percent among revolving-credit respondents, 79.3 percent among other-borrowing respondents, and 85.1 percent among respondents unable to pay $400. These estimates are descriptive adult-respondent associations, not causal effects or validated forecasts. The contribution is a measurement scaffold for distinguishing liquidity sufficiency, liquidity bridges, debt-financed distress, and liquidity exclusion in household financial-resilience research.

1. Introduction

Household financial resilience is often summarized by income, emergency savings, or debt. Each measure is useful, but each is incomplete. Income describes a flow of resources. Liquid buffers describe immediate shock absorption. Credit can be a payment bridge, a sign of distress, or unavailable when needed. Hardship outcomes such as bill pressure, external help, and medical debt measure realized strain. A measurement framework that collapses these margins into a single statistic misses important distinctions.

This paper uses the 2025 SHED public-use microdata to study those distinctions. The empirical objective is deliberately modest. The paper does not estimate the causal effect of savings or credit on hardship. Instead, it asks how hardship outcomes are associated with liquid-buffer status and emergency-financing profiles in a single adult-respondent cross-section.

The main innovation is a profile decomposition. SHED respondents can report several ways they would handle a $400 emergency expense. A respondent who would pay with a credit card and pay it in full is not in the same economic state as a respondent who would revolve a credit-card balance, borrow from other sources, be unable to pay, or appear unable to use credit despite low buffers. The paper therefore builds a priority-coded emergency-financing taxonomy and then tests how sensitive the resulting profiles are to alternative priority rules and multiple-response behavior.

The central descriptive pattern is sharp. Using a strict hardship outcome that excludes credit constraint, the financial-wellbeing summary, and the unable-to-pay-$400 item, hardship is lowest among bridge-credit-paid-in-full respondents and much higher among respondents reporting revolving credit, other emergency borrowing, inability to pay, or constrained non-use. This pattern survives basic income and demographic adjustment, but it should still be interpreted as descriptive clustering rather than causal evidence.

The paper is best read as a field or policy-measurement draft. Its purpose is to make household-resilience measurement more precise by separating liquidity sufficiency, liquidity bridges, debt-financed distress, and liquidity exclusion. The revised draft adds non-exclusive response-dummy validation, pooled small-cell treatment, benchmark-screen comparisons, multi-year stability checks using harmonized SHED variables for 2015–2025, and bill-pressure validation for 2023–2025. A stronger journal version should still add correct SHED survey-design inference, data-driven classification, and a fuller literature review.

2. Conceptual frame

The motivating concept is household resilience: the ability to avoid hardship after a shock. In causal form, one might define resilience as a lower probability of hardship conditional on a shock and household characteristics: resilience equals one minus the probability of hardship after conditioning on the shock, household covariates, and macro context.

The 2025 SHED cross-section used here does not identify that causal estimand. There is no clean exogenous shock timing, and the key variables are contemporaneous. The empirical object in this draft is therefore an association between hardship and income, liquid buffers, emergency-financing profile, and household controls.

The paper avoids predictive language except in the loose measurement sense of variables containing information about hardship in the same cross-section. If future versions use the language of prediction, they should implement holdout validation, calibration, and benchmark prediction metrics.

Four latent states guide interpretation:

1. **Liquidity sufficient:** the respondent can use cash, checking, savings, or similar liquid resources.
2. **Liquidity bridge:** the respondent uses credit as short-term payment smoothing and can pay it in full.
3. **Debt-financed distress:** the respondent would revolve credit or use other emergency borrowing.
4. **Liquidity exclusion:** the respondent cannot pay, lacks listed financing, or appears constrained despite low buffers.

These states are not welfare rankings. They are a measurement vocabulary for separating different ways households report handling a small emergency expense.

3. Data, population, and weights

The analysis uses the 2025 Survey of Household Economics and Decisionmaking public-use microdata. The unit of observation is an adult respondent. Some questions refer to household circumstances, but the weighted population in this draft should be read as weighted adult respondents, not directly as a count of households. The main analytic sample contains 12,932 adult respondents with nonmissing controls and positive weights in the benchmark models, representing about 264.5 million weighted adults.

The paper uses SHED final weights for weighted descriptive estimates and weighted linear-probability models. The public-use files inspected for this draft include final cross-sectional weights and population weights but not public replicate weights, strata, or PSU variables. Reported confidence intervals for proportions therefore use an approximate effective-sample calculation, and model standard errors are approximate robust WLS/LPM standard errors. These are useful for scale and screening, but they are not final design-consistent SHED survey inference. A journal version should either obtain or implement official design-variance guidance if available, or present this public-use limitation explicitly.

The draft builds from a replication scaffold that downloads public SHED files, constructs analysis variables, and records source provenance. For a public replication package, the necessary ingredients are: public-use file source, variable names, construction code, weighting code, sample restrictions, missing-data treatment, and generated tables. This manuscript summarizes the constructs; a separate appendix should provide codebook-level detail.

4. Construct definitions and overlap controls

Table 1. SHED hardship-measurement construct map. Revised construct definitions. The primary outcome is a strict non-overlap hardship measure that excludes the emergency-capacity item, credit constraint, and the broad financial-wellbeing summary.

Construct	Variable	Definition	Role	Caveat
Low liquid buffer	low_buffer_both	Cannot cover $400 with cash/equivalent and lacks three months emergency savings	Exposure	Not a bank-balance measure; SHED response-based
Credit reliance	credit_reliance_any	Would pay $400 by credit over time or another emergency borrowing response	Moderator/exposure	Broad first-pass emergency-financing screen; excludes credit-constraint outcome to reduce mechanical overlap
Emergency-financing profile	emergency_financing_profile	Mutually exclusive profile: unable to pay, revolving credit, other borrowing, constrained non-use, bridge credit paid in full, cash/checking/savings, or other	Decomposition	Priority-coded descriptive taxonomy; not a welfare ranking
Primary strict hardship	hardship_strict_nonoverlap	Bill pressure, external help, or medical debt	Primary outcome	Excludes credit constraint, financial-wellbeing summary, and unable-pay-$400
Broad hardship composite	hardship_composite	Financial vulnerability, bill pressure, external help, credit constraint, or medical debt	Robustness outcome	Excludes unable-pay-$400 but still overlaps with constrained-nonuse profile through credit constraint
Bill pressure	bill_pressure	Did not pay all non-card bills in full or had difficult bills	Outcome	2025 build; harmonization needed for multi-year work
External help	fs21_any_external_help	Received outside-household help with listed expenses	Outcome	Descriptive support reliance
Medical debt	medical_debt	Currently has debt from medical care	Outcome	Links to MEPS lane but remains SHED self-report

Table 1 separates exposures, profiles, and outcomes. The low-buffer exposure is strict: a respondent both cannot cover a $400 emergency with cash or equivalent and lacks three months of emergency savings. Emergency credit reliance is an initial broad screen for revolving credit or other borrowing in response to a $400 emergency.

The revised draft changes the primary outcome. The earlier broad composite included credit constraint, which created a mechanical-overlap problem for constrained non-use profiles. The primary outcome is now **strict non-overlap hardship**: bill pressure, external help with expenses, or medical debt. It excludes the $400 inability item, credit constraint, and the broad financial-wellbeing summary. Broader composites are still reported, but they are no longer the headline outcome.

Table 2. Sample and overlap diagnostics for SHED 2025 constructs. Weighted respondent-population estimates. These diagnostics clarify which outcomes are broad composites and which are designed to avoid mechanical overlap with emergency-capacity and credit-constraint items.

Construct variable	Observed N	Weighted adult pop.	Weighted rate % [95% approx CI]
low_buffer_both	12,932	264.5M	30.4 [29.6, 31.3]
credit_reliance_any	12,934	264.5M	25.3 [24.5, 26.1]
hardship_strict_nonoverlap	12,934	264.5M	45.5 [44.6, 46.4]
hardship_no_credit_constraint	12,934	264.5M	50.6 [49.7, 51.5]
hardship_composite	12,934	264.5M	53.5 [52.5, 54.4]
financially_vulnerable_b2	12,934	264.5M	27.1 [26.3, 27.9]
bill_pressure	12,934	264.5M	27.8 [27.0, 28.6]
fs21_any_external_help	12,934	264.5M	23.2 [22.4, 24.0]
credit_constrained	12,934	264.5M	18.6 [17.9, 19.4]
medical_debt	12,934	264.5M	18.3 [17.6, 19.0]
unable_pay_400	12,934	264.5M	12.2 [11.6, 12.8]

Table 2 documents the observed denominators and weighted adult rates. The strict non-overlap outcome is intentionally narrower than the broad composite. This lowers the risk that profile definitions mechanically generate the outcome, especially for constrained non-use.

5. Buffer-credit profiles

Table 3. Hardship by liquid-buffer and credit-reliance profile, SHED 2025. Weighted descriptive rates. The primary strict outcome excludes credit constraint, financial-wellbeing summary, and unable-pay-$400 to reduce mechanical overlap.

Profile	N	Weighted pop.	Strict hardship % [95% approx CI]	Broad composite %	Financially vulnerable %	Bill pressure %	Medical debt %
Higher buffer / no credit reliance	8,530	169.9M	28.0 [27.0, 29.1]	34.9	9.9	11.0	10.5
Higher buffer / credit reliance	666	14.1M	64.5 [60.6, 68.4]	71.9	29.5	38.2	30.1
Low buffer / no credit reliance	1,289	27.6M	83.4 [81.2, 85.6]	93.0	70.2	70.5	33.0
Low buffer / credit reliance	2,447	52.9M	76.9 [75.0, 78.7]	87.6	59.1	56.9	32.7

Table 3 provides a simple two-by-two profile: higher versus low buffer, and no emergency credit reliance versus emergency credit reliance. Strict hardship is much more common among low-buffer respondents. It is also higher among higher-buffer respondents who report emergency credit reliance than among higher-buffer respondents without that reliance.

The low-buffer/no-credit-reliance group remains especially important. Its strict hardship rate is high, which suggests that non-use of emergency credit is not necessarily a sign of resilience. Some non-users may be liquidity excluded or constrained. This is why the more detailed emergency-financing profile is necessary.

6. Emergency-financing profile decomposition

Table 4. Pooled emergency-financing profiles and hardship, SHED 2025. Small diagnostic cells are pooled into `Liquidity exclusion / unclassified` for main-text interpretation. The detailed priority-coded table remains available as a diagnostic appendix table. Profiles are descriptive groups, not causal treatments or welfare rankings.

Emergency-financing profile	N	Weighted pop.	Strict hardship % [95% approx CI]	Broad composite %	Low buffer %	Financially vulnerable %	Bill pressure %	Credit constrained %	Medical debt %
Cash/checking/savings	3,754	76.2M	31.4 [29.8, 33.0]	39.7	0.5	12.0	12.9	11.7	13.0
Bridge credit paid in full	4,704	92.1M	24.7 [23.4, 26.0]	30.3	0.3	7.8	8.7	6.2	8.1
Revolving credit for $400	1,833	38.3M	67.9 [65.6, 70.2]	80.3	74.4	47.6	45.9	30.8	32.4
Other emergency borrowing	943	21.5M	79.3 [76.5, 82.1]	87.3	82.7	52.8	56.0	36.1	28.2
Unable to pay $400	1,530	32.3M	85.1 [83.2, 87.0]	93.5	92.7	73.4	72.9	42.7	35.3
Liquidity exclusion / unclassified	170	4.1M	78.8 [72.2, 85.4]	86.8	90.0	48.2	62.8	32.3	27.1

Table 4 is the main measurement table. It separates bridge credit from distress-oriented financing while pooling very small diagnostic cells into a broader liquidity-exclusion/unclassified category for main-text interpretation. Bridge credit paid in full has a strict hardship rate of 24.7 percent. Cash/checking/savings has a strict hardship rate of 31.4 percent. In contrast, revolving credit for $400 has a strict hardship rate of 67.9 percent, other emergency borrowing 79.3 percent, inability to pay 85.1 percent, and the pooled liquidity-exclusion/unclassified category 78.8 percent.

The bridge-credit result should be interpreted carefully. It is not evidence that paid-in-full credit causes lower hardship. It likely marks a group with stronger liquidity management, better credit access, higher unobserved financial sophistication, or other advantages. The useful point is measurement: paid-in-full credit and revolving credit should not be combined into a single credit-use category.

The pooled liquidity-exclusion/unclassified category combines constrained non-use, low-buffer/no-listed-financing, and other/unclassified respondents. This is a conservative presentation choice: the detailed categories remain diagnostically useful, but constrained non-use and other/unclassified are too small in the 2025 build to serve as stable headline groups.

7. Multiple-response validation and priority-rule sensitivity

Table 5. Common non-exclusive $400 emergency-response combinations, SHED 2025. Respondents can select multiple response options. This table validates why any mutually exclusive profile requires an explicit priority rule.

Response combination	N	Weighted pop.	Strict hardship % [95% approx CI]	Broad composite %
cash/savings	3,761	76.4M	31.5 [29.9, 33.0]	39.8
card paid full	3,633	71.2M	24.9 [23.4, 26.4]	30.3
unable	1,161	24.5M	83.3 [81.0, 85.6]	92.8
cash/savings + card paid full	1,075	21.0M	24.2 [21.4, 26.9]	30.8
card over time	1,034	21.0M	69.2 [66.2, 72.3]	82.4
other borrowing	645	14.8M	80.9 [77.6, 84.1]	89.8
cash/savings + card over time	227	4.6M	60.2 [53.5, 67.0]	75.5
other borrowing + unable	210	4.5M	93.0 [89.3, 96.7]	96.6
card over time + other borrowing	202	4.5M	80.4 [74.5, 86.4]	91.4
none/listed other	159	3.8M	79.4 [72.6, 86.2]	85.8
cash/savings + other borrowing	160	3.7M	80.1 [73.3, 86.9]	88.0
cash/savings + card paid full + card over time	131	2.9M	50.8 [41.5, 60.1]	60.6

SHED respondents can select multiple $400 emergency response options. Table 5 shows the most common non-exclusive combinations. The table validates the central coding issue: any mutually exclusive profile requires an explicit priority rule. Many respondents select a single dominant response, but meaningful combinations exist, including cash plus paid-in-full card, cash plus revolving credit, other borrowing plus inability to pay, and card-over-time plus other borrowing.

Table 6. Emergency-financing profile sensitivity to priority rule. The primary rule prioritizes inability to pay and distress borrowing before bridge/cash options. The alternative rule prioritizes bridge credit paid in full and cash before revolving/borrowing. Differences show where multiple-response behavior affects classification.

Profile	Primary N	Primary pop.	Primary strict hardship %	Bridge-first N	Bridge-first pop.	Bridge-first strict hardship %
Cash/checking/savings	3,754	76.2M	31.4	4,243	86.9M	36.1
Bridge credit paid in full	4,704	92.1M	24.7	5,121	101.2M	28.1
Revolving credit for $400	1,833	38.3M	67.9	1,236	25.5M	71.2
Other emergency borrowing	943	21.5M	79.3	645	14.8M	80.9
Unable to pay $400	1,530	32.3M	85.1	1,530	32.3M	85.1
Constrained non-use	51	1.2M	85.5	40	0.9M	90.0
Low buffer / no listed financing	100	2.5M	76.4	100	2.5M	76.4
Other or unclassified	19	0.4M	74.1	19	0.4M	74.1

Table 6 compares the primary priority rule with an alternative bridge-first rule. The primary rule prioritizes inability to pay and distress borrowing before bridge/cash options. The alternative rule prioritizes paid-in-full credit and cash before revolving/borrowing. The main qualitative contrast remains: bridge and cash profiles are lower-hardship than revolving, other borrowing, inability to pay, and constrained/non-use profiles. But the population assigned to cash and bridge groups changes, and their hardship rates rise under bridge-first coding because some multi-response respondents are pulled out of distress categories.

This sensitivity check is important. It shows that the taxonomy is promising but not final. The next section adds non-exclusive response dummy models and a first-pass data-driven clustering validation. A stronger submission version should still test alternative clustering specifications, multiple-correspondence analysis, or latent class analysis.

8. Adjusted descriptive validation

Table 7. Emergency-financing profile coefficients for strict hardship, SHED 2025. Weighted LPM coefficients in percentage points relative to `Cash/checking/savings`, with approximate robust standard errors in parentheses. Controls include income, education, age, race/ethnicity, housing tenure, employment, children, and region. N=12,932; weighted adult population=264.5M; R^2=0.297.

Profile relative to cash/checking/savings	Coefficient pp (SE)	CI low	CI high
Bridge credit paid in full	-2.9 (1.0)	-4.9	-0.9
Constrained non-use	40.5 (5.3)	30.1	51.0
Low buffer / no listed financing	28.7 (4.3)	20.3	37.0
Other emergency borrowing	33.8 (1.7)	30.5	37.2
Other or unclassified	32.3 (10.6)	11.6	53.1
Revolving credit for $400	29.2 (1.5)	26.3	32.1
Unable to pay $400	37.4 (1.5)	34.5	40.4

Table 7 reports an adjusted weighted linear-probability model for strict hardship, using cash/checking/savings as the reference category. Relative to that group, bridge credit paid in full is associated with 2.9 percentage points lower strict hardship. Revolving credit is associated with 29.2 points higher strict hardship, other emergency borrowing with 33.8 points higher, inability to pay with 37.4 points higher, and the small diagnostic categories are elevated but imprecise. The profile coefficients are more central to the paper than the model-sequence R^2 values because they directly test the measurement distinction between bridge credit and distress-oriented financing.

Table 8. Non-exclusive emergency-response dummy model for strict hardship, SHED 2025. Weighted LPM coefficients in percentage points with approximate robust standard errors in parentheses. Emergency-response indicators are not mutually exclusive. Controls include income, education, age, race/ethnicity, housing tenure, employment, children, and region. N=12,932; weighted adult population=264.5M; R^2=0.295. Descriptive validation benchmark, not causal inference.

Non-exclusive response / exposure	Coefficient pp (SE)	CI low	CI high
Paid-in-full card response	-7.6 (1.2)	-9.9	-5.3
Revolving card response	11.9 (1.4)	9.2	14.7
Cash/checking/savings response	-5.5 (1.1)	-7.7	-3.4
Other emergency borrowing response	13.3 (1.4)	10.6	16.0
Unable-to-pay response	14.2 (1.5)	11.2	17.1
Low buffer	14.2 (1.6)	11.1	17.4

Table 8 adds non-exclusive emergency-response dummy validation. Because SHED respondents can select more than one $400 emergency response, this model does not force every respondent into a single priority-coded profile. The pattern supports the taxonomy. Paid-in-full card and cash/checking/savings responses are negatively associated with strict hardship after controls, while revolving card, other borrowing, inability to pay, and low-buffer status are positively associated with strict hardship. This reduces the concern that the main result is only an artifact of the priority rule.

9. Data-driven taxonomy validation

Table 9. Data-driven emergency-financing clusters, SHED 2025. Weighted k-means clusters use emergency-response, low-buffer, and credit-constraint indicators only; hardship outcomes are held out and reported after clustering. Cluster labels are interpretive summaries of centroid rates, not new causal classes.

Cluster	Interpretive label	N	Weighted pop.	Strict hardship % [95% approx CI]	Financially vulnerable %	Bill pressure %	Cash response %	Paid-full card %	Revolving card %	Other borrowing %	Unable-to-pay %	Low buffer %	Credit constrained %
C1	Empirical bridge-credit cluster	3,719	72.9M	25.7 [24.2, 27.2]	8.8	10.2	0.0	99.6	1.1	0.9	0.1	0.3	6.6
C2	Empirical cash-plus-bridge cluster	1,269	25.2M	29.5 [26.8, 32.2]	8.7	8.9	100.0	100.0	10.7	6.3	0.5	7.5	7.2
C3	Empirical cash/liquid-sufficiency cluster	3,910	79.5M	32.9 [31.3, 34.5]	12.8	13.8	99.8	0.0	2.8	1.3	0.3	0.5	12.2
C4	Empirical revolving / borrowing cluster	2,576	56.0M	74.8 [72.9, 76.6]	53.7	54.3	15.7	6.4	60.3	48.7	1.8	87.6	35.5
C5	Empirical unable-to-pay / exclusion cluster	1,458	30.9M	84.9 [82.9, 86.9]	73.6	72.8	2.6	0.4	3.4	16.0	100.0	93.8	42.5

Table 9 is a first-pass data-driven validation exercise. The clusters use emergency-response, low-buffer, and credit-constraint indicators only; strict hardship, financial vulnerability, and bill pressure are held out until after cluster assignment. This is not a latent-class model and should not be oversold. It is a transparent robustness check asking whether the response pattern itself separates liquid sufficiency, bridge credit, revolving/borrowing, and inability-to-pay/exclusion states.

The result is encouraging. The empirical bridge-credit cluster has strict hardship of 25.7 percent, and the cash-plus-bridge cluster has 29.5 percent. The cash/liquid-sufficiency cluster is somewhat higher at 32.9 percent, while the revolving/borrowing cluster is much higher at 74.8 percent and the unable-to-pay/exclusion cluster is 84.9 percent. Because outcomes were not used in clustering, this ordering strengthens the claim that the taxonomy reflects structure in emergency-financing responses rather than only author-imposed outcome sorting.

Table 10. Crosswalk from priority-coded profiles to data-driven clusters, SHED 2025. Cells show the weighted percent of each pooled priority-coded profile assigned to each empirical cluster. This is a robustness diagnostic: high concentration supports the author-imposed taxonomy, while dispersion identifies mixed-response profiles.

Priority-coded pooled profile	C1	C2	C3	C4	C5
Cash/checking/savings	0.0	0.0	100.0	0.0	0.0
Bridge credit paid in full	77.2	22.8	0.0	0.0	0.0
Revolving credit for $400	2.0	6.9	5.8	85.3	0.0
Other emergency borrowing	3.1	6.6	3.2	87.2	0.0
Unable to pay $400	0.1	0.4	0.8	3.2	95.6
Liquidity exclusion / unclassified	7.7	1.5	3.4	87.3	0.0

Table 10 crosswalks the priority-coded pooled profiles to the empirical clusters. Cash/checking/savings maps cleanly to the empirical cash cluster. Bridge credit paid in full mostly maps to the empirical bridge cluster, with a smaller cash-plus-bridge group. Revolving credit, other emergency borrowing, and inability to pay also concentrate in the expected empirical clusters. The pooled liquidity-exclusion/unclassified category is more dispersed, which is exactly why the paper treats it as a pooled diagnostic category rather than a precise headline class.

10. Benchmark screens

Table 11. Benchmark screens for strict hardship, SHED 2025. Weighted descriptive comparison of simpler screens against the pooled emergency-financing profile taxonomy. Positive/negative rates are unadjusted strict-hardship rates. Model R^2 is from controls plus the listed screen or profile indicators and is included only as secondary descriptive validation.

Screen / profile set	Positive N / model N	Positive pop. / model pop.	Strict hardship if positive %	Strict hardship if negative %	Unadjusted gap pp	Controls + screen/profile R^2
Income below $50k	4,342	88.7M	69.7	33.4	36.3	0.219
Cannot cover $400 with cash/equivalent	4,534	97.5M	76.4	27.5	49.0	0.294
Lacks three months emergency savings	5,606	119.6M	68.1	26.9	41.1	0.266
Low buffer: both $400 and 3-month savings	3,736	80.5M	79.1	30.8	48.3	0.280
Broad emergency credit reliance	3,115	67.0M	74.3	35.8	38.5	0.260
Pooled emergency-financing profile	12,932	264.5M	—	—	—	0.296

Table 11 benchmarks the profile taxonomy against simpler screens. The comparison is deliberately modest: each screen is useful, and the table is not a prediction exercise. Income below $50,000, inability to cover $400 with cash/equivalent, lacking three months of emergency savings, the combined low-buffer measure, and broad credit reliance all separate higher-hardship from lower-hardship respondents. The pooled emergency-financing profile adds a different kind of information: it does not merely identify who is exposed, but separates cash sufficiency, paid-in-full bridge credit, revolving credit, other borrowing, inability to pay, and liquidity exclusion.

The profile model's R^2 is close to the $400 cash/equivalent and low-buffer screens, but the paper's argument does not rest on a small R^2 ranking. The main value is interpretive. A single low-buffer or credit-reliance flag cannot show whether credit use is a bridge, distress borrowing, or unavailable when needed.

Table 12. Benchmark model sequence for SHED 2025 strict hardship. Weighted linear-probability models with approximate robust standard errors in parentheses. Controls include income, education, age, race/ethnicity, housing tenure, employment, children, and region. Descriptive, not causal. R^2 is secondary validation context, not the main evidence.

Model	N	Weighted adult pop.	Low buffer pp	Credit reliance pp	Low buffer × credit pp	R^2
M1	12,932	264.5M	—	—	—	0.219
M2	12,932	264.5M	31.7 (1.1)	—	—	0.280
M3	12,932	264.5M	25.2 (1.3)	10.8 (1.2)	—	0.286
M4	12,932	264.5M	35.7 (1.5)	26.8 (2.0)	-27.8 (2.5)	0.294

Table 12 reports a simpler model sequence. Adding low-buffer status to income and demographic controls raises R^2 from 0.219 to 0.280 for strict hardship. Adding emergency credit reliance raises R^2 to 0.286, and adding the interaction raises R^2 to 0.294. These in-sample R^2 changes are secondary. They are not presented as validation of a prediction model; they are descriptive evidence that buffers and emergency-financing behavior contain information not captured by income and basic demographics.

A proper prediction exercise would require train/test splits or cross-validation, calibration, and benchmark metrics such as AUC, Brier score, or log loss. This draft does not make that claim.

11. Component outcomes

Table 13. Low-buffer and credit-reliance interactions across hardship outcomes, SHED 2025. Weighted LPM coefficients in percentage points with approximate robust standard errors. Each row uses the same controls as Table 12. Descriptive clustering, not causal effects.

Outcome	N	Weighted adult pop.	Low buffer pp	Credit reliance pp	Low buffer × credit pp	R^2
hardship_strict_nonoverlap	12,932	264.5M	35.7 (1.5)	26.8 (2.0)	-27.8 (2.5)	0.294
hardship_no_credit_constraint	12,932	264.5M	37.6 (1.4)	27.7 (1.9)	-27.4 (2.3)	0.342
hardship_composite	12,932	264.5M	35.5 (1.3)	26.2 (1.9)	-25.2 (2.2)	0.339
financially_vulnerable_b2	12,932	264.5M	43.8 (1.7)	13.9 (1.9)	-19.1 (2.6)	0.348
bill_pressure	12,932	264.5M	45.3 (1.7)	20.6 (2.0)	-29.8 (2.7)	0.312
fs21_any_external_help	12,932	264.5M	6.4 (1.7)	13.0 (1.9)	-3.7 (2.7)	0.184
credit_constrained	12,932	264.5M	21.1 (1.7)	7.7 (1.7)	-5.5 (2.5)	0.150
medical_debt	12,932	264.5M	22.2 (1.6)	18.1 (2.0)	-19.5 (2.6)	0.106

Table 13 reports the low-buffer, credit-reliance, and interaction specification across multiple outcomes. This table is included because the composite should not carry the argument alone. Low buffers are strongly associated with financial vulnerability, bill pressure, credit constraints, and medical debt. Emergency credit reliance is also positively associated with the strict composite and with component outcomes. The interaction is negative in several rows, likely reflecting outcome saturation and compositional differences among low-buffer respondents.

The component grid supports the paper's main caution: emergency credit reliance is not a single mechanism. The same broad credit indicator can mix bridge credit, revolving debt, other borrowing, inability to pay, and constrained non-use. Future work should use the profile taxonomy rather than only the broad binary credit-reliance variable.

12. Multi-year validation

The revised draft adds repeated-cross-section validation using the harmonized SHED panel. The multi-year file supports a simplified emergency-financing taxonomy from 2015 through 2025: cash/checking/savings, bridge credit paid in full, revolving credit for $400, unable to pay $400, low-buffer/no-listed-financing, and other/unclassified. It does not contain the full 2025 strict composite because external help and medical debt are not harmonized across the full period. The validation therefore uses two outcomes: financial vulnerability for 2015–2025 and bill pressure for 2023–2025.

Table 14. Multi-year stability: simplified emergency-financing profiles and financial vulnerability, SHED 2015–2025. Harmonized repeated cross-sections. This longer-window validation uses the financial-vulnerability outcome because strict bill-pressure coverage is available only in 2023–2025.

Year	Profile	N	Financial vulnerability % [95% approx CI]
2015	Cash/checking/savings	1,759	16.3 [14.0, 18.5]
2015	Bridge credit paid in full	1,522	10.7 [8.7, 12.7]
2015	Revolving credit for $400	948	44.5 [40.3, 48.7]
2015	Unable to pay $400	852	72.7 [68.6, 76.7]
2015	Low buffer / no listed financing	472	57.2 [51.1, 63.2]
2015	Other or unclassified	89	29.6 [17.0, 42.2]
2016	Cash/checking/savings	1,551	18.5 [15.9, 21.2]
2016	Bridge credit paid in full	2,396	9.4 [7.8, 10.9]
2016	Revolving credit for $400	1,265	44.9 [41.3, 48.6]
2016	Unable to pay $400	876	75.6 [71.7, 79.6]
2016	Low buffer / no listed financing	453	51.5 [45.1, 57.8]
2016	Other or unclassified	69	27.8 [13.4, 42.3]
2017	Cash/checking/savings	3,669	13.8 [12.3, 15.3]
2017	Bridge credit paid in full	4,245	8.2 [7.1, 9.4]
2017	Revolving credit for $400	2,068	42.8 [40.0, 45.6]
2017	Unable to pay $400	1,408	69.6 [66.4, 72.8]
2017	Low buffer / no listed financing	661	52.1 [47.1, 57.2]
2017	Other or unclassified	136	22.6 [13.4, 31.7]
2018	Cash/checking/savings	3,679	12.3 [11.0, 13.5]
2018	Bridge credit paid in full	3,589	6.8 [5.8, 7.7]
2018	Revolving credit for $400	1,707	43.6 [40.6, 46.5]
2018	Unable to pay $400	1,434	68.3 [65.2, 71.3]
2018	Low buffer / no listed financing	744	48.0 [43.5, 52.6]
2018	Other or unclassified	163	29.9 [20.9, 38.8]
2019	Cash/checking/savings	4,158	11.9 [10.7, 13.0]
2019	Bridge credit paid in full	4,222	6.6 [5.8, 7.5]
2019	Revolving credit for $400	1,758	41.5 [38.8, 44.2]
2019	Unable to pay $400	1,360	69.8 [66.9, 72.7]
2019	Low buffer / no listed financing	611	56.1 [51.3, 61.0]
2019	Other or unclassified	64	23.8 [10.7, 36.8]
2020	Cash/checking/savings	3,945	11.9 [10.8, 13.0]
2020	Bridge credit paid in full	4,249	6.9 [6.1, 7.7]
2020	Revolving credit for $400	1,627	43.1 [40.5, 45.6]
2020	Unable to pay $400	1,245	71.9 [69.2, 74.6]
2020	Low buffer / no listed financing	524	56.9 [52.2, 61.6]
2020	Other or unclassified	58	23.2 [11.3, 35.2]
2021	Cash/checking/savings	4,206	12.3 [11.2, 13.4]
2021	Bridge credit paid in full	4,531	6.0 [5.3, 6.8]
2021	Revolving credit for $400	1,534	40.8 [38.1, 43.6]
2021	Unable to pay $400	1,089	70.5 [67.4, 73.6]
2021	Low buffer / no listed financing	443	53.1 [47.7, 58.4]

Year	Profile	N	Financial vulnerability % [95% approx CI]
2021	Other or unclassified	71	29.2 [16.9, 41.5]
2022	Cash/checking/savings	3,737	12.9 [11.7, 14.1]
2022	Bridge credit paid in full	4,282	8.7 [7.7, 9.6]
2022	Revolving credit for $400	1,705	45.6 [43.0, 48.3]
2022	Unable to pay $400	1,367	70.9 [68.2, 73.6]
2022	Low buffer / no listed financing	504	61.1 [56.3, 65.9]
2022	Other or unclassified	72	43.7 [30.2, 57.1]
2023	Cash/checking/savings	3,669	15.3 [14.0, 16.5]
2023	Bridge credit paid in full	4,116	8.7 [7.7, 9.6]
2023	Revolving credit for $400	1,687	47.2 [44.6, 49.9]
2023	Unable to pay $400	1,337	74.5 [71.9, 77.1]
2023	Low buffer / no listed financing	518	59.7 [54.9, 64.4]
2023	Other or unclassified	73	32.7 [20.7, 44.7]
2024	Cash/checking/savings	3,811	13.8 [12.6, 14.9]
2024	Bridge credit paid in full	4,630	9.2 [8.3, 10.1]
2024	Revolving credit for $400	1,724	47.4 [44.8, 49.9]
2024	Unable to pay $400	1,419	69.8 [67.2, 72.4]
2024	Low buffer / no listed financing	609	59.2 [54.8, 63.5]
2024	Other or unclassified	102	39.2 [28.9, 49.6]
2025	Cash/checking/savings	3,922	13.8 [12.7, 15.0]
2025	Bridge credit paid in full	4,846	8.3 [7.5, 9.1]
2025	Revolving credit for $400	1,833	47.6 [45.1, 50.1]
2025	Unable to pay $400	1,530	73.4 [71.0, 75.8]
2025	Low buffer / no listed financing	714	60.8 [56.9, 64.6]
2025	Other or unclassified	89	28.3 [18.3, 38.3]

Table 14 shows that the broad profile ordering is stable across the longer window. Bridge credit paid in full is consistently the lowest-vulnerability or near-lowest-vulnerability profile. Revolving credit is much higher, inability to pay is highest, and low-buffer/no-listed-financing is also elevated. The exact levels move across years, but the taxonomy is not a one-year artifact.

Table 15. Multi-year validation: simplified emergency-financing profiles and bill pressure, SHED 2023–2025. Bill pressure is the cleanest harmonized strict-hardship component available in the harmonized panel for 2023–2025.

Year	Profile	N	Bill pressure % [95% approx CI]
2023	Cash/checking/savings	3,669	8.2 [7.2, 9.2]
2023	Bridge credit paid in full	4,116	3.9 [3.3, 4.6]
2023	Revolving credit for $400	1,687	21.4 [19.3, 23.6]
2023	Unable to pay $400	1,337	48.8 [45.8, 51.8]
2023	Low buffer / no listed financing	518	39.8 [35.1, 44.5]
2023	Other or unclassified	73	25.6 [14.4, 36.7]
2024	Cash/checking/savings	3,811	7.8 [6.9, 8.7]
2024	Bridge credit paid in full	4,630	4.5 [3.9, 5.1]
2024	Revolving credit for $400	1,724	19.8 [17.8, 21.9]
2024	Unable to pay $400	1,419	46.0 [43.2, 48.9]
2024	Low buffer / no listed financing	609	41.4 [37.1, 45.8]
2024	Other or unclassified	102	25.2 [16.1, 34.4]
2025	Cash/checking/savings	3,922	14.6 [13.4, 15.8]
2025	Bridge credit paid in full	4,846	9.4 [8.5, 10.3]
2025	Revolving credit for $400	1,833	45.9 [43.5, 48.4]
2025	Unable to pay $400	1,530	72.9 [70.5, 75.3]
2025	Low buffer / no listed financing	714	65.6 [61.9, 69.3]
2025	Other or unclassified	89	43.8 [32.8, 54.9]

Table 15 uses bill pressure, the cleanest strict-hardship component available in the harmonized panel for 2023–2025. The pattern again separates bridge/cash profiles from revolving credit, inability to pay, and low-buffer/no-listed-financing. Bill pressure rises sharply in 2025 across profiles, but the ordering remains informative: bridge credit paid in full and cash/checking/savings remain far below revolving credit and inability to pay.

Table 16. Multi-year benchmark model sequence. Weighted LPM in-sample R^2 for validation only. These are not prediction metrics; they test whether low-buffer and profile variables add descriptive information beyond controls and year fixed effects.

Outcome/window	Model	N	R^2
2023–2025 bill pressure	Income/demographics + year FE	36,627	0.176
2023–2025 bill pressure	+ low-buffer indicator	36,627	0.238
2023–2025 bill pressure	+ emergency-financing profile	36,627	0.261
2015–2025 financial vulnerability	Income/demographics + year FE	60,168	0.206
2015–2025 financial vulnerability	+ low-buffer indicator	60,168	0.318
2015–2025 financial vulnerability	+ emergency-financing profile	60,168	0.335

Table 16 reports multi-year benchmark model sequences. These are still in-sample descriptive models, not prediction exercises. For 2023–2025 bill pressure, adding low-buffer status raises R^2 from 0.176 to 0.238, and adding emergency-financing profile raises it to 0.261. For 2015–2025 financial vulnerability, adding low-buffer status raises R^2 from 0.206 to 0.318, and adding profile raises it to 0.335. The repeated-cross-section evidence supports the measurement value of the taxonomy, while still leaving survey-design inference and external validation as future work.

13. Interpretation

The revised evidence supports a narrower claim than the first draft. It does not show that liquid buffers or credit behavior cause hardship. It does not establish a validated household-resilience index. It does show that SHED emergency-financing responses can be organized into economically meaningful profiles and that those profiles are strongly associated with non-overlapping hardship outcomes.

The most useful distinction is between liquidity bridges and distress financing. Paid-in-full credit resembles a bridge profile. Revolving credit, other borrowing, inability to pay, and constrained non-use resemble distress or exclusion profiles. Cash/checking/savings sits in between in this draft, partly because many households with liquid resources can still have bills pressure, external support needs, or medical debt.

This structure is a better measurement object than a single credit-use variable. It also offers a clearer empirical path for the broader household-resilience research agenda: define profiles carefully, validate them across years, compare them against simpler screens, and only then consider whether they improve prediction or policy classification.

14. Limitations

Five limitations are central.

First, the analysis is cross-sectional and descriptive. The results are contemporaneous associations among adult respondents.

Second, survey inference is incomplete. The current intervals and standard errors are approximate and should be replaced with the correct SHED variance procedure before journal submission.

Third, the profile taxonomy is priority-coded. The sensitivity tables, non-exclusive response-dummy model, pooled small-cell presentation, first-pass clustering validation, benchmark screens, and multi-year validation help, but richer data-driven classification methods should still be explored.

Fourth, small cells should not be overread. Constrained non-use and other/unclassified profiles are useful diagnostics but not stable headline categories in the 2025 build.

Fifth, the literature positioning remains a work in progress. A submission version should engage directly with buffer-stock saving, liquidity constraints, wealthy hand-to-mouth households, financial fragility, consumer credit access, and the Federal Reserve's SHED reporting.

15. Next steps

The next empirical upgrade should focus on validation rather than rhetoric:

1. implement SHED design-consistent variance estimation;
2. extend the full 2025 strict-composite construction as far as possible across comparable SHED years;
3. test additional emergency-financing priority rules and pooled small-cell specifications;
4. extend the first-pass clustering validation with multiple-correspondence analysis, latent class analysis, and out-of-sample classification checks;
5. evaluate profile classification against additional income-only, emergency-savings-only, and $400-capacity-only benchmarks;
6. if prediction language is retained, add cross-validation, AUC, Brier score, calibration, and benchmark comparisons;
7. write a full data appendix with variable names, coding, missingness, weights, and sample restrictions.

16. Conclusion

This revised draft turns the broad household-resilience idea into a narrower SHED measurement exercise. The paper's contribution is not a causal estimate and not a final resilience index. Its contribution is a clearer taxonomy: liquidity sufficiency, liquidity bridges, debt-financed distress, and liquidity exclusion should be measured separately.

The bridge-credit versus distress-borrowing distinction is the strongest result. Paid-in-full credit and revolving credit look very different in the SHED data, and inability to pay is different again. Treating all of these states as generic credit use obscures the structure of household hardship.

The paper is therefore best developed as a field or policy-measurement article. The revised non-exclusive response model, first-pass data-driven clustering validation, pooled small-cell presentation, benchmark-screen comparison, and multi-year evidence strengthen the case that the profile ordering is not only a 2025 priority-coding artifact. The remaining validation path is clear: design-consistent inference, data-driven taxonomy checks, and deeper literature engagement.

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